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GATE ECE · COMMUNICATIONS · CHAPTER-WISE PYQS

Fundamentals of Information Theory

GATE ECE Communications · Previous Year Questions with Answer Key

32	24	1989 – 2026	English
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Marks 2	20 questions
Answer key	at the end

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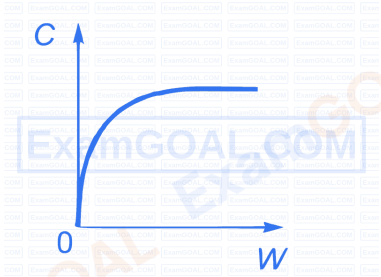
Marks 1 12 questions

Q1. GATE ECE 2025

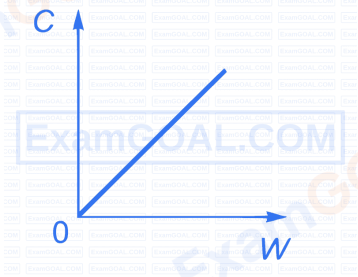
Open solution

Consider an additive white Gaussian noise (AWGN) channel with bandwidth W and noise power spectral density $\frac{N_o}{2}$. Let P_{av} denote the average transmit power constraint. Which one of the following plots illustrates the dependence of the channel capacity C on the bandwidth W (keeping P_{av} and N_0 fixed)?

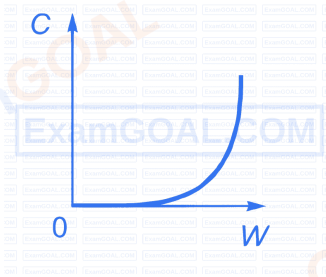
(A)



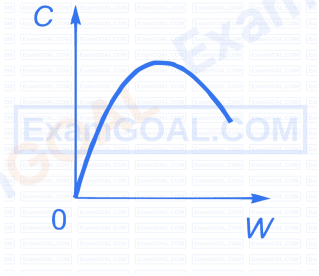
(B)



(C)



(D)



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Q2. GATE ECE 2025

The generator matrix of a $(6, 3)$ binary linear block code is given by

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

The minimum Hamming distance $d_{\min}$ between codewords equals_____ (answer in integer).

Open solution

Q3. GATE ECE 2024

A source transmits symbols from an alphabet of size 16. The value of maximum achievable entropy (in bits) is _____ .

Open solution

Q4. GATE ECE 2021

A speech signal, band limited to 4 kHz , is sampled at 1.25 times the Nyquist rate. The speech samples, assumed to be statistically independent and uniformly distributed in the range -5 V to +5 V , are subsequently quantized in an $8-$ bit uniform quantizer and then transmitted over a voice - grade AWGN telephone channel. If the ratio of transmitted signal power to channel noise power is 26 dB , the minimum channel bandwidth required to ensure reliable transmission of the signal with arbitrarily small probability of transmission error (rounded off to two decimal places) is ____ kHz .

Open solution

Q5. GATE ECE 2020

A binary random variable X takes the value +2 or -2 . The probability $P(X = +2) = \alpha$. The value of α (rounded off to one decimal place), for which the entropy of X is maximum, is ____ .

Open solution

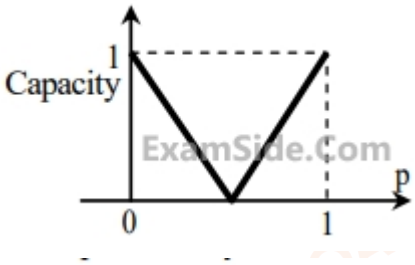
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Q6. GATE ECE 2017 Set 2

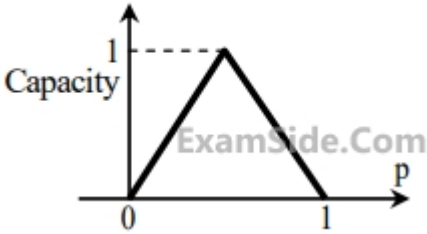
Which one of the following graphs shows the Shannon capacity (channel capacity) in bits of a memory less binary symmetric channel with crossover probability P?

Open solution

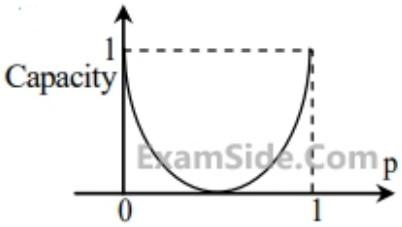
(A)



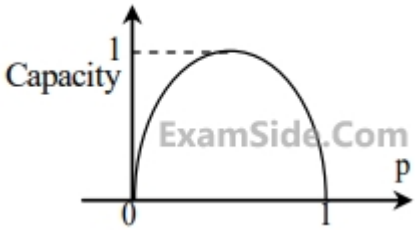
(B)



(C)



(D)



Q7. GATE ECE 2017 Set 1

Let (X_1, X_2) be independent random variables, X_1 has mean 0 and variance 1, while X_2 has mean 1 and variance 4. The mutual information I (X_1, X_2) between X_1 and X_2 in bits is _____.

Open solution

Q8. GATE ECE 2016 Set 2

A discrete memoryless source has an alphabet (a_1, a_2, a_3, a_4) with corresponding probabilities$\left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\right)$. The minimum required average codeword length in bits to represent this source for error-free reconstruction is_____

Open solution

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Q9. GATE ECE 2016 Set 3

An analog baseband signal, band limited to 100 Hz, is sampled at the Nyquist rate. The samples are quantized into four message symbols that occur independently with probabilities p_1 = p_4 = 0.125 and p_2 =p_3. The information rate (bits/sec) of the message source is _____

Open solution

Q10. GATE ECE 2012

A source alphabet consists of N symbols with the probability of the first two symbols being the same. A source encoder increases the probability of the first symbol by a small amount ε and decreases that of the second by ε. After encoding, the entropy of the source

(A) increases
(B) remains the same
(C) increases only if N = 2
(D) decreases

Open solution

Q11. GATE ECE 2012

In a base band communications link, frequencies up to 3500 Hz are used for signaling. Using a raised cosine pulse with 75% excess bandwidth and for no inter-symbol interference, the maximum possible signaling rate in symbols per second is

(A) 1750 (B) 2625 (C) 4000 (D) 5250

Open solution

Q12. GATE ECE 2011

An analog signal is band-limited to 4kHz. sampled at the Nyquist rate and the samples are quantized into 4 levels. The quantized levels are assumed to be independent and equally probable, bit rate is

(A) 1 bit/sec (B) 2 bits/sec (C) 3 bits/sec (D) 4 bits/sec

Open solution

Marks 2 20 questions

Q13. GATE ECE 2026

Let the relevant bandwidth (B) of a digital communication system be 1 MHz and $kT = -174\mathrm{dBm/Hz}$, where k is Boltzmann's constant and ' T ' is equivalent noise temperature of the receiver. The power (S) of signal received through an additive Gaussian channel is -80 dBm .

Which of the following options is/are TRUE about Shannon capacity (C) of the channel?

(A) $C = B$ (B) $C = 2B$ (C) $C > 3B$ (D) $C < B$

Open solution

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Q14. GATE ECE 2025

Open solution

The random variable X takes values in $\{-1, 0, 1\}$ with probabilities $P(X = -1) = P(X = 1) = \alpha$ and $P(X = 0) = 1 - 2\alpha$, where $0 < \alpha < \frac{1}{2}$.

Let $g(\alpha)$ denote the entropy of X (in bits), parameterized by α. Which of the following statements is/are TRUE?

(A) $g(0.4) > g(0.3)$ **(B)** $g(0.3) > g(0.4)$ **(C)** $g(0.3) > g(0.25)$ **(D)** $g(0.25) > g(0.3)$

Q15. GATE ECE 2025

Open solution

X and Y are Bernoulli random variables taking values in $\{0, 1\}$. The joint probability mass function of the random variables is given by:

$P(X = 0, Y = 0) = 0.06$
$P(X = 0, Y = 1) = 0.14$
$P(X = 1, Y = 0) = 0.24$
$P(X = 1, Y = 1) = 0.56$

The mutual information $I(X; Y)$ is _____(rounded off to two decimal places).

Q16. GATE ECE 2023

Open solution

The frequency of occurrence of 8 symbols (a-h) is shown in the table below. A symbol is chosen and it is determined by asking a series of "yes/no" questions which are assumed to be truthfully answered. The average number of questions when asked in the most efficient sequence, to determine the chosen symbol, is _____ (rounded off to two decimal places).

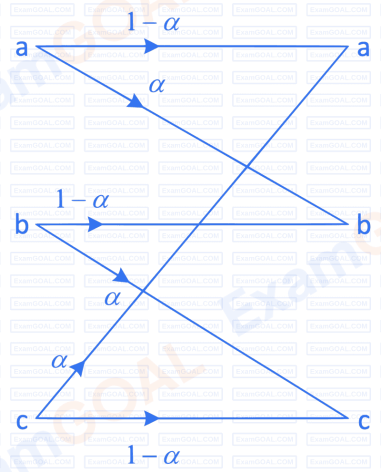
Symbols	a	b	c	d	e	f	g	h
Frequency of occurrence	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{128}$	$\frac{1}{128}$

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Q17. GATE ECE 2022

Open solution

The transition diagram of a discrete memoryless channel with three input symbols and three output symbols is shown in the figure. The transition probabilities are as marked. The parameter α lies in the interval [0.25, 1]. The value of .. for which the capacity of this channel is maximized, is _____ (rounded off to two decimal places).



Q18. GATE ECE 2022

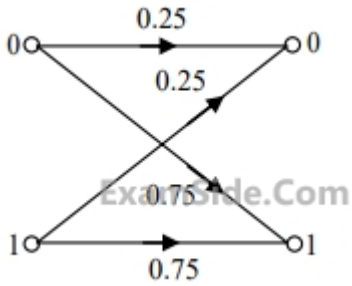
Open solution

Consider communication over a memoryless binary symmetric channel using a (7, 4) Hamming code. Each transmitted bit is received correctly with probability $(1 - \in)$, and flipped with probability $\in$. For each codeword transmission, the receiver performs minimum Hamming distance decoding, and correctly decodes the message bits if and only if the channel introduces at most one bit error. For $\in$ = 0.1, the probability that a transmitted codeword is decoded correctly is _____ (rounded off to two decimal places).

Q19. GATE ECE 2017 Set 2

Open solution

Consider a binary memoryless channel characterized by the transition probability diagram shown in the figure.



The channel is

(A) Lossless (B) Noiseless (C) Useless (D) Deterministic

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Q20. GATE ECE 2016 Set 2

Open solution

A binary communication system makes use of the symbols “zero” and “one”. There are channel errors. Consider the following events:

x_0 : a " zero " is transmitted

x_1 : a " one " is transmitted

y_0 : a " zero " is received

y_1 : a " one " is received

The following probabilities are given:

$P(x_0) = \frac{3}{4}, P(y_0|x_0) = \frac{1}{2}, and P(y_0|x_1) = \frac{1}{2}.$

The information in bits that you obtain when you learn which symbol has been received (while you know that a " zero " has been transmitted) is _____

Q21. GATE ECE 2016 Set 1

Open solution

Consider a discreet memoryless source with alphabet $S = \{s_0, s_1, s_2, s_3, s_{4.....}\}$ and respective probabilities of occurrence $P = \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, ...\}$. The entropy of the source (in bits) is_____.

Q22. GATE ECE 2016 Set 3

Open solution

A voice-grade AWGN (additive white Gaussian noise) telephone channel has a bandwidth of 4.0 kHz and two-sided noise power spectral density $\frac{\eta}{2} = 2.5 \times 10^{-5}$ Watt per Hz. If information at the rate of 52 kbps is to be transmitted over this channel with arbitrarily small bit error rate, then the minimum bit energy E_b (in mJ/bit) necessary is _____

Q23. GATE ECE 2014 Set 1

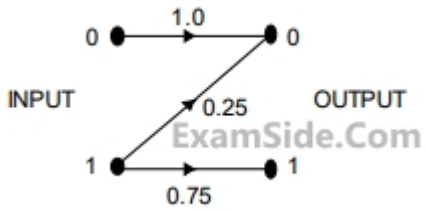
Open solution

A fair coin is tossed repeatedly until a 'Head' appears for the first time. Let L be the number of tosses to get this first 'Head'. The entropy H (L) in bits is _____.

Q24. GATE ECE 2014 Set 4

Open solution

Consider the Z- channel given in the figure. The input is 0 or 1 with equal probability.



If the output is 0, the probability that the input is also 0 equals_____

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Q25. GATE ECE 2014 Set 2

The capacity of band-limited additive white Gaussian noise (AWGN) channel is given by $C = W \log_2 \left(1 + \frac{P}{\sigma^2 W}\right)$ bits per second (bps), where W is the channel bandwidth, P is the average power received and σ^2 is the one-sided power spectral density of the AWGN.
For a Fixed $\frac{P}{\sigma^2} = 1000$, the channel capacity (in kbps) with infinite band width $(W \to \infty)$ is approximately

(A) 1.44 (B) 1.08 (C) 0.72 (D) 0.36

Open solution

Q26. GATE ECE 2009

A communication channel with AWGN operating at a signal to noise ratio SNR > > 1 and band width B has capacity C_1 . If the SNR is doubled keeping B constant, the resulting capacity C_2 is given

(A) $C_2 \approx 2\,C_1$ (B) $C_2 \approx C_1 + B$
(C) $C_2 \approx C_1 + 2B$ (D) $C_2 \approx C_1 + 0.3B$

Open solution

Q27. GATE ECE 2008

A memory less source emits n symbols each with a probability p. The entropy of the source as a function of n

(A) increases as log n (B) decreases as log (1 / n)
(C) increases as n (D) increases as n log n

Open solution

Q28. GATE ECE 2006

A source generates three symbols with probabilities 0.25, 0.25, 0.50 at a rate of 3000 symbols per second. Assuming independent generation of symbols, the most efficient source encoder would have average bit rate as

(A) 6000 bits/sec (B) 4500 bits/sec (C) 3000 bits/sec (D) 1500 bits/sec

Open solution

Q29. GATE ECE 2001

A video transmission system transmits 625 picture frames per second. Each frame consists of a 400×400 pixel grid with 64 intensity levels per pixel. The data rate of the system is

(A) 16 Mbps (B) 100 Mbps (C) 600 Mbps (D) 6.4 Gbps

Open solution

Q30. GATE ECE 1991

A binary source has symbol probabilities 0.8 and 0.2. If extension coding (blocks of 4 symbols) is used, the lower and upper bounds on the average code word length are :
(a)lower_____.
(b) higher_____.

Open solution

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Q31. GATE ECE 1990

An image uses 512×512 picture elements. Each of the picture elements can take any of the 8 distinguishable intersity levels. The maximum entropy in the above image will be

(A) 2097152 bits (B) 786432 bits (C) 648 bits (D) 144 bits

Open solution

Q32. GATE ECE 1989

A source produces 4 symbols with probabilities $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$ and $\frac{1}{8}$. For this source, a practical coding scheme has an average codeword lenght of 2 bits/symbols. The efficiency of the code is :

(A) 1 (B) 7/8 (C) 1/2 (D) 1/4

Open solution

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Answer Key

Marks 1

1. **A**	2. **3**	3. **4**	4. **9.2 to 9.3**	5. **0.5**
6. **C**	7. **0**	8. **1.75**	9. **362.2**	10. **D**
11. **C**	12. **D**			

Marks 2

13. **C**	14. **B, C**	15. **0**	16. **1.97 to 1.99**	17. **1**
18. **0.85**	19. **C**	20. **0.82**	21. **2**	22. **31.5**
23. **2**	24. **0.8**	25. **A**	26. **B**	27. **A**
28. **B**	29. **C**	30. **(a) 2.8 (b) 3.8**	31. **B**	32. **B**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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